

The quantum chromatic number of the G_{19} join family

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Abstract

Let G_{19} be the 19-vertex complex orthogonality graph introduced by Lalonde and put $J_n = G_{19} \vee K_{n-3}$ for $n \geq 3$. Lalonde proved the claimed value for color projectors of rank one [2, Theorem 1.5] and conjectured the same value without a rank restriction [2, discussion following Theorem 1.5]. We prove that conjecture in the finite-dimensional projector model:

$$\chi_q(J_n) = n + 1 \quad (n \geq 3).$$

In particular, adjoining one apex to G_{19} gives a 20-vertex graph H with $\chi_q(H) = 5$.

The proof first symmetrizes an arbitrary coloring, so that every projector has a common rank r . For a fixed color, a rational sum-of-squares identity rigidifies vertices $1, \dots, 13$ into a rank- r sign-vector configuration. The remaining six subspaces are classified, including all non-transverse branches, by an r -plane invariant under a canonical complex structure. Orthogonality between two colors then preserves the two complex-structure sectors while reversing membership in the invariant planes. The resulting sector subspaces are pairwise orthogonal across colors, and their dimensions give the impossible inequality $3nr \leq 2nr$. The core sum-of-squares and all formal block identities have an exact, standard-library verifier.

1 Graph, model, and result

The vertex set of G_{19} is $\{1, \dots, 19\}$, and its edge set is

$$\begin{aligned} E_{19} = \{ & 12, 13, 18, 19, 1\,15, 23, 26, 27, 2\,14, 34, 35, 3\,16, 45, 4\,12, 4\,13, \\ & 5\,10, 5\,11, 5\,17, 67, 6\,11, 6\,13, 7\,10, 7\,12, 7\,19, 89, 8\,11, 8\,12, 8\,18, \\ & 9\,10, 9\,13, 14\,17, 14\,18, 15\,17, 15\,19, 16\,18, 16\,19\}. \end{aligned}$$

Here vw denotes the unordered edge (v, w) . Its graph6 representation is

$$\text{RxLAKA@AgYAWDGO?O?@??A?W@00C@_}.$$

The only triangles are

$$T_0 = 123, \quad T_1 = 189, \quad T_2 = 345, \quad T_3 = 267.$$

Here $\vee$ denotes graph join and K_0 is the empty graph. For $n \geq 3$, define

$$J_n = G_{19} \vee K_{n-3}.$$

Thus the graph in the question is $H = J_4$, with vertex 20 as the joined apex.

An m -coloring in the finite-dimensional projector model is a family of orthogonal projections $P_{v,c} \in M_d(\mathbb{C})$, with zero projections allowed, such that

$$\sum_{c=1}^m P_{v,c} = I_d, \quad P_{v,c} P_{w,c} = 0 \quad (vw \in E).$$

The row orthogonality $P_{v,c} P_{v,c'} = 0$ for $c \neq c'$ follows from, and may equivalently be included with, the assertion that the row is a projective partition of the identity. By the structure theorem recorded in [2, Theorem 2.19], following Cameron–Montanaro–Newman–Severini–Winter [1], this projector formulation is equivalent to the standard finite-dimensional definition of χ_q by perfect entangled strategies for the graph-coloring game.

Theorem 1.1 (Main theorem). *For every $n \geq 3$,*

$$\chi_q(G_{19} \vee K_{n-3}) = n + 1.$$

In particular, $\chi_q(H) = 5$.

For the elementary upper bound, G_{19} has the following four-coloring:

v	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\kappa(v)$	1	2	3	2	1	3	1	2	3	2	4	3	1	1	2	1	3	3	3

Giving the $n-3$ joined vertices fresh colors proves $\chi(J_n) \leq n+1$. The work is to exclude a quantum n -coloring. This also excludes every coloring with fewer than n colors, since unused colors may be padded by zero projections.

2 Clique saturation and uniformization

Lemma 2.1 (Clique columns). *Suppose an m -coloring contains an m -clique C . Then, for every color c ,*

$$\sum_{v \in C} P_{v,c} = I_d.$$

Thus the $m \times m$ array $(P_{v,c})_{v \in C, c \in [m]}$ is a magic unitary.

Proof. For fixed c , the summands are pairwise orthogonal, so their sum is a projection of trace at most d . Summing this trace bound over the m columns gives at most md . The row identities show that the total trace is exactly md , so equality holds in every column. A projection of trace d in $M_d(\mathbb{C})$ is the identity. $\square$

Lemma 2.2 (Color uniformization). *If J_n has a finite-dimensional quantum n -coloring, then it has one in a dimension $D = nr$ for which every projector has the same positive rank r .*

Proof. Given a coloring in dimension d , take its direct sum over the $n!$ color permutations:

$$\tilde{P}_{v,c} = \bigoplus_{\pi \in S_n} P_{v,\pi(c)}.$$

This is again a coloring. Its dimension is $D = n!d$, while

$$\text{rank } \tilde{P}_{v,c} = (n-1)! \sum_{a=1}^n \text{rank } P_{v,a} = (n-1)!d =: r.$$

Hence $D = nr$ and $r > 0$. $\square$

Assume henceforth that such a uniform coloring exists. Fix a color c . Let R_c be the sum of the color- c projections on the joined clique K_{n-3} and let

$$Q_c = I_D - R_c.$$

The joined vertices are pairwise adjacent, so $\text{rank } R_c = (n-3)r$ and $\dim Q_c = 3r$. Every old-vertex projection $P_{v,c}$ is supported on Q_c . Applying Lemma 2.1 to $T_j \vee K_{n-3}$ gives

$$\sum_{v \in T_j} P_{v,c} = Q_c \quad (j = 0, 1, 2, 3). \quad (1)$$

Thus, on Q_c , the fixed-color projections form a uniform rank- r projective orthogonal representation of G_{19} in dimension $3r$.

3 Exact fixed-color rigidity

In this section fix the color and write $e_v = P_{v,c}|_{Q_c}$ and $I = I_{Q_c}$. All traces are the ordinary matrix trace on Q_c .

3.1 A rational core sum of squares

Put

$$X = e_{10} + e_{11} + e_{12} + e_{13}$$

and define the three Walsh forms

$$\begin{aligned} H_1 &= e_{10} + e_{11} - e_{12} - e_{13}, \\ H_2 &= e_{10} - e_{11} + e_{12} - e_{13}, \\ H_3 &= e_{10} - e_{11} - e_{12} + e_{13}. \end{aligned}$$

Also put $d_1 = e_5 - e_4$, $d_2 = e_7 - e_6$, and $d_3 = e_9 - e_8$.

Lemma 3.1 (Core SOS). *The defining projection, edge, and triangle relations imply the exact identity*

$$\frac{4}{3}I - X = \sum_{j=0}^3 f_j^* f_j, \quad (2)$$

where all f_j are self-adjoint and

$$f_0 = \frac{1}{2} \left(X - \frac{4}{3}I \right), \quad f_i = \frac{2}{3}d_i + \frac{1}{2}H_i \quad (1 \leq i \leq 3).$$

Proof. The triangle decompositions and edge zeros give

$$\sum_{i=1}^3 d_i^2 = (e_4 + e_5) + (e_6 + e_7) + (e_8 + e_9) = 2I. \quad (3)$$

Hadamard cancellation, together with $e_j^2 = e_j$, gives

$$X^2 + H_1^2 + H_2^2 + H_3^2 = 4X. \quad (4)$$

Finally,

$$\sum_{i=1}^3 \{d_i, H_i\} = -4X. \quad (5)$$

To see the last identity, collect the coefficient of each e_j for $j = 10, 11, 12, 13$. The corresponding signed sum of d_1, d_2, d_3 is $N_j - (2I - N_j) = 2N_j - 2I$, where N_j is the sum of the three projections among $e_4, \dots, e_9$ adjacent to j . Since $N_j e_j = e_j N_j = 0$, its anticommutator with e_j is $-4e_j$. Summing over j proves (5).

Expanding the right side of (2) and inserting (3)–(5) leaves exactly $\frac{4}{3}I - X$. $\square$

The two sides of (2) have the same trace, namely zero, because $\dim Q_c = 3r$ and each e_j has rank r . Faithfulness of the matrix trace therefore forces every f_j to vanish. Define

$$A = e_4 - e_5, \quad B = e_6 - e_7, \quad C = e_8 - e_9.$$

Walsh inversion now gives

$$\begin{aligned} e_{10} &= \frac{1}{3}(I + A + B + C), & e_{11} &= \frac{1}{3}(I + A - B - C), \\ e_{12} &= \frac{1}{3}(I - A + B - C), & e_{13} &= \frac{1}{3}(I - A - B + C). \end{aligned} \quad (6)$$

Moreover,

$$A^2 = I - e_3, \quad B^2 = I - e_2, \quad C^2 = I - e_1, \quad A^2 + B^2 + C^2 = 2I.$$

Inserting (6) into $e_j^2 = e_j$ and separating the four Walsh sign patterns yields

$$\{B, C\} = A, \quad \{A, C\} = B, \quad \{A, B\} = C. \quad (7)$$

Lemma 3.2 (Core normal form). *There is a Hilbert space K_c of dimension r and a unitary $W_c : K_c^3 \rightarrow Q_c$ under which the ranges of $e_1, \dots, e_{13}$ are*

$$\begin{aligned} 1 &= (I, 0, 0), & 2 &= (0, I, 0), & 3 &= (0, 0, I), \\ 4 &= (I, I, 0), & 5 &= (I, -I, 0), \\ 6 &= (I, 0, I), & 7 &= (I, 0, -I), \\ 8 &= (0, I, I), & 9 &= (0, I, -I), \\ 10 &= (I, I, I), & 11 &= (I, I, -I), \\ 12 &= (I, -I, I), & 13 &= (I, -I, -I). \end{aligned}$$

Here a block row denotes the range of the associated block-column map from K_c ; normalization of the frame is immaterial to its range.

Proof. Use $e_1 + e_2 + e_3 = I$ to write $Q_c = K_1 \oplus K_2 \oplus K_3$, with $K_i = \text{ran } e_i$ and $\dim K_i = r$. The operator A is supported on $K_1 \oplus K_2$, B on $K_1 \oplus K_3$, and C on $K_2 \oplus K_3$. In the first relation of (7), the (1, 1) and (2, 2) blocks of $\{B, C\}$ vanish, so the two diagonal blocks of A vanish. Cyclically, all diagonal blocks vanish. Since $A^2 = I - e_3$, its (1, 2) block is unitary, and similarly for B, C . Thus

$$A = \begin{pmatrix} 0 & U & 0 \\ U^* & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & V \\ 0 & 0 & 0 \\ V^* & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & W \\ 0 & W^* & 0 \end{pmatrix}.$$

Equation (7) gives $U = VW^*$, $V = UW$, and $W = U^*V$. Conjugating by the block-diagonal unitary $\text{diag}(I, U^*, V^*)$ makes all three off-diagonal blocks the identity. The binary decompositions and (6) then give the displayed ranges. $\square$

3.2 All tail branches

The tail is not always the graph of an operator; retaining degenerate planes is essential.

Lemma 3.3 (Tail-plane classification). *After the normalization in Lemma 3.2, the ranges at vertices $14, \dots, 19$ are determined by an r -plane $M_c \subset K_c \oplus K_c$ satisfying*

$$J_c M_c = M_c, \quad J_c = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}. \quad (8)$$

Explicitly,

$$\begin{aligned} 14 &= \{(a, 0, b) : (a, b) \in M_c\}, & 15 &= \{(0, a, b) : (a, b) \in M_c\}, \\ 16 &= \{(a, -b, 0) : (a, b) \in M_c\}, & & \\ 17 &= \{(x, x, z) : (x, z) \in M_c^\perp\}, & 18 &= \{(x, -z, z) : (x, z) \in M_c^\perp\}, \\ 19 &= \{(x, -z, x) : (x, z) \in M_c^\perp\}. \end{aligned} \quad (9)$$

Conversely, every J_c -invariant r -plane gives rank- r subspaces satisfying all fixed-color edge relations.

Proof. The six core–tail edges 2 14, 1 15, 3 16, 5 17, 8 18, and 7 19 place the tail ranges in the images of the following injective two-column frames:

v	$F_v(a, b)$	v	$F_v(a, b)$
14	$(a, 0, b)$	17	(a, a, b)
15	$(0, a, b)$	18	$(a, -b, b)$
16	$(a, -b, 0)$	19	$(a, -b, a)$

Write $N_v \subset K_c^2$ for the r -dimensional parameter plane at v . For the six tail–tail edges, direct pullback gives

vw	14 17	15 17	14 18	16 18	16 19	15 19
$F_v^* F_w$	I	I	I	I	I	$-J_c$

Put $M_c = N_{14}$. The first five entries successively give $N_{17} = M_c^\perp$, $N_{15} = M_c$, $N_{18} = M_c^\perp$, $N_{16} = M_c$, and $N_{19} = M_c^\perp$. The final entry says $J_c M_c^\perp = M_c^\perp$. Because J_c is unitary and $J_c^* = -J_c$, this is equivalent to $J_c M_c = M_c$. The displayed frames are exactly (9). To make the complement equalities explicit, every edge first gives an orthogonality inclusion between two parameter planes. Each plane has dimension r in the same $2r$ -dimensional space, as does the orthogonal complement on the right, so every such inclusion is an equality. Reversing these equalities verifies the converse. $\square$

Remark 3.4. *If the first-coordinate projection of M_c is invertible, then $M_c = \text{ran}(I, S)$ and (8) is $S^2 = -I$. This transverse branch recovers the familiar operator-valued normal form. It is not exhaustive. For instance, $M = \text{span}\{(e_1, 0), (0, e_1)\} \subset \mathbb{C}^2 \oplus \mathbb{C}^2$ is J -invariant but is not a graph. Lemma 3.3 includes this branch.*

4 Cross-color sector obstruction

Take distinct colors c, d and use the core isometries from Lemma 3.2. Put

$$T_{dc} = W_d^* W_c : K_c^3 \longrightarrow K_d^3.$$

Lemma 4.1 (Cross-color block equations). *There are maps $X, Y, Z : K_c \rightarrow K_d$ such that*

$$T_{dc} = \begin{pmatrix} 0 & X & Y \\ -X & 0 & Z \\ -Y & -Z & 0 \end{pmatrix}. \quad (10)$$

For $L : K_c \rightarrow K_d$, define

$$\Omega_L = \begin{pmatrix} 0 & L \\ -L & 0 \end{pmatrix}.$$

Then, for each $L \in \{X, Y, Z\}$,

$$\Omega_L M_c \subseteq M_d^\perp, \quad \Omega_L M_c^\perp \subseteq M_d. \quad (11)$$

Proof. Same-vertex orthogonality at vertices 1, 2, 3 kills the diagonal blocks of T_{dc} . Vertices 4, 6, 8 make the corresponding lower block the negative of the upper block, giving (10). No adjoints occur here: T_{dc} maps the coordinate space for c to the one for d .

Compressing (10) by the two-column frames in (9), vertices 14, 15, 16 give, respectively,

$$\Omega_Y M_c \subseteq M_d^\perp, \quad \Omega_Z M_c \subseteq M_d^\perp, \quad -\Omega_X M_c \subseteq M_d^\perp.$$

Vertices 17, 18, 19 give

$$\Omega_{Y+Z} M_c^\perp \subseteq M_d, \quad \Omega_{Y-X} M_c^\perp \subseteq M_d, \quad \Omega_{Z-X} M_c^\perp \subseteq M_d.$$

The coefficient transform is invertible; for example

$$2X = (Y + Z) - (Y - X) - (Z - X),$$

with analogous formulas for Y, Z . This proves (11). $\square$

Let $\Gamma_c = 2P_{M_c} - I$ on K_c^2 . Equation (11) is equivalent to

$$\Gamma_d \Omega_L = -\Omega_L \Gamma_c \quad (L = X, Y, Z). \quad (12)$$

Because M_c reduces the unitary J_c , it splits across the $+i$ and $-i$ eigenspaces of J_c . Using the isometries up to a harmless factor

$$V_{c,\sigma}(x) = (x, \sigma i x), \quad \sigma \in \{+, -\},$$

write

$$M_c \cap \ker(J_c - \sigma i I) = V_{c,\sigma}(E_{c,\sigma}), \quad e_{c,\sigma} = \dim E_{c,\sigma}.$$

Then

$$e_{c,+} + e_{c,-} = r. \quad (13)$$

The operator Ω_L intertwines J_c with J_d and acts within a σ sector as the nonzero scalar σi times L . Combining this with (12) gives

$$L E_{c,\sigma} \subseteq E_{d,\sigma}^\perp \quad (c \neq d, L = X, Y, Z). \quad (14)$$

We can now finish the proof.

Proof of Theorem 1.1. Suppose that J_n had a quantum n -coloring. Uniformize it using Lemma 2.2, so the physical dimension is $D = nr$ and all ranks are $r > 0$. For each color c , apply Lemmas 3.2 and 3.3, and define the physical sector subspace

$$\mathcal{H}_{c,\sigma} = W_c(E_{c,\sigma} \oplus E_{c,\sigma} \oplus E_{c,\sigma}).$$

It has dimension $3e_{c,\sigma}$. Equations (10) and (14) show the pairwise orthogonality directly. For $c \neq d$, take $\xi = (\xi_1, \xi_2, \xi_3) \in E_{c,\sigma}^{\oplus 3}$ and $\eta = (\eta_1, \eta_2, \eta_3) \in E_{d,\sigma}^{\oplus 3}$. Then

$$\begin{aligned} \langle W_d \eta, W_c \xi \rangle &= \langle \eta, T_{dc} \xi \rangle \\ &= \langle \eta_1, X\xi_2 + Y\xi_3 \rangle + \langle \eta_2, -X\xi_1 + Z\xi_3 \rangle \\ &\quad + \langle \eta_3, -Y\xi_1 - Z\xi_2 \rangle = 0. \end{aligned}$$

Every term vanishes by (14): $\eta_j \in E_{d,\sigma}$, whereas $X\xi_k, Y\xi_k, Z\xi_k \in E_{d,\sigma}^\perp$. Thus, for fixed σ , the n subspaces $\mathcal{H}_{c,\sigma}$ are pairwise orthogonal in the common nr -dimensional physical space. Therefore

$$3 \sum_{c=1}^n e_{c,+} \leq nr, \quad 3 \sum_{c=1}^n e_{c,-} \leq nr.$$

Adding and using (13) gives

$$3nr = 3 \sum_{c=1}^n (e_{c,+} + e_{c,-}) \leq 2nr,$$

which is impossible because $n, r > 0$. Hence no quantum n -coloring exists. The explicit classical $(n+1)$ -coloring given in Section 1 proves equality. $\square$

5 Relation to the rank-one theorem and exact verification

The previously known rank-one argument classifies scalar three-dimensional orthogonal representations of G_{19} . The proof above does not assume that classification and does not assume equal ranks in the original coloring. Instead, color symmetrization makes the fixed-color ranks uniform; the core SOS classifies the resulting fusion-frame core; and Lemma 3.3 treats every higher-rank tail, including non-graph and noncommuting branches. Thus the obstruction applies to arbitrary finite dimension, zero projectors, nonuniform original ranks, reducible representations, and irreducible representations alike. The induced subgraph on vertices $1, \dots, 13$ is the Mančinska–Roberson graph G_{13} . Their operator propagation proving $\chi_q(G_{13}) = 4$ [3] is therefore a relevant earlier higher-rank rigidity precedent; the SOS and exhaustive invariant-plane tail classification here use the additional structure of the present 19-vertex graph.

The short verifier `verify_obstruction_certificate.py` reads `obstruction_certificate.json` and replays the paper’s four-factor SOS, neighbor signs, six tail compressions, J intertwining, and terminal dimension coefficients. Independently, `verify_lalonde_uniform_obstruction.py` reads `lalonde_uniform_obstruction.json`, uses a distinct trace-SOS formulation, and additionally checks graph6 decoding, the clique-ideal witness, and row reduction of the cross-color kernel. A separate `verify_graph.py` checks the canonical graph data. All arithmetic is exact and rational; no numerical SDP output enters the theorem.

Artifact provenance. The exact algebraic artifacts audited before these presentation revisions are frozen in an immutable project snapshot with full Git object name

b1944a23707eb69d2f9f25eda0bb73c32cd5500a.

Relative to that project root, the certificate payloads have SHA-256 digests

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certificate/obstruction_certificate.json:  
fdf2aff0df2902b08ad0363e6448e299c7b04d1437b329634c9d7a719376c76c  
certificate/lalonde_uniform_obstruction.json:  
e5e0a6126c1adc4088e090ea58ca194f8c2f0ce243f9cd9c4c151329e97a71e1
```

The final source commit and rendered-PDF digest are release metadata recorded after typesetting is frozen; omitting them here avoids a self-referential artifact hash.

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References

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